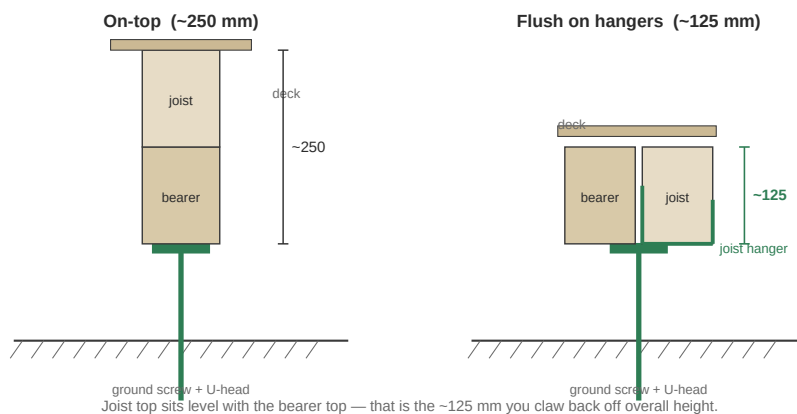


Timber workshops — ground-screw foundation & floor plan

LOW PROFILE — flush joists on hangers, ~125 mm floor

Chosen to keep **overall building height** to a minimum. Joists frame in **flush** with the bearers on joist hangers, so the whole floor is one ~125 mm layer instead of a ~250 mm stack — roughly **125 mm lower** to the top of deck. Ground-screw count stays low; the cost is the hangers and the cutting. Same 5"×2" timber. Prepared 30 Jul 2026.

Where the height goes



Design assumptions

- **Joists:** 5"×2" (47×125 mm) C24, at **400 mm centres**, framing **flush** into the bearers on **joist hangers** (top faces level).
- **Bearers:** 5×2 on the ground-screw heads, at **2.0 m centres**; screws at ≤1.5 m along each.
- Joists are **cut at every bearer** and hung both faces, so each is a **simple span** ≤ **2.0 m** (no continuity — keep C24, and for heavy floor loads tighten bearers to 1.5 m).
- Fixings: hangers nailed with sherardised twist nails; bearers coach-screwed to the U-heads.
- Deck: 18–22 mm OSB3 T&G, laid flush over joists and bearers.

Headline quantities

- **Ground screws:** ~33 total — 15 for Workshop 1, ~18 for Workshop 2 (order ~36 with spares).
- **5×2 timber:** ~200 lin. m of framing — ~230 m with waste (same order as the on-top build).
- **Joist hangers:** ~135 (47×125 timber-to-timber) — the item that buys you the low floor.

Workshop 1 — 6.0 × 4.0 m

- **Ground screws:** 15 — 5 columns × 3 bearer lines (Y = 0, 2.0, 4.0 m).
- **Bearers:** 3 × 6.0 m = **18 m**. **Joists:** 16 lines × 4.0 m = **64 m** (each cut into two 2.0 m bays).
- **Joist hangers:** 16 joists × 4 seats = ~64.

Workshop 2 — ~4.0 × 7.2 m (clipped corner)

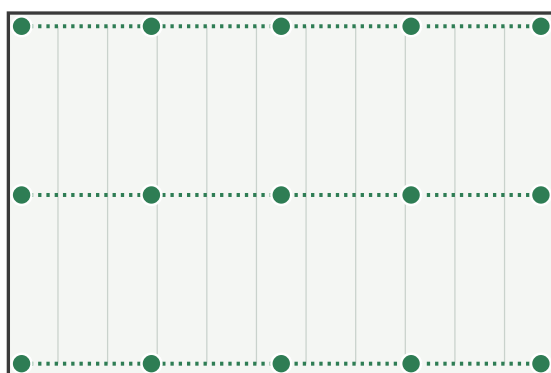
- **Ground screws:** ~18 — 3 bearer lines (X = 0, 2.0, 4.0 m); screws at Y = 0, 1.5, 3.0, 4.5, 6.0, 7.2 m, plus 2 at the door threshold.
- **Bearers:** ≈ **21 m**. **Joists:** ~19 lines × 4.0 m ≈ **72–76 m**.
- **Joist hangers:** ~19 joists × 4 seats ≈ ~70 (a few fewer at the clip).

Shopping list

Item	Quantity	Notes
5x2 joists	35 × 4.2 m	one length per joist line, cut into 2.0 m bays
5x2 bearers + noggins	~13 × 4.8 m	bearers on the screws + a little blocking
5x2 total	≈ 48 lengths	~230 lin. m incl. waste
Ground screws	36 (33 + 3 spares)	adjustable U-head / plate top
Joist hangers	~140	47×125 timber-to-timber, galvanised
Hanger / twist nails	~2000 (2 tubs)	sherardised 3.75 × 30 square twist
Coach screws	~70	bearers down to the screw U-heads

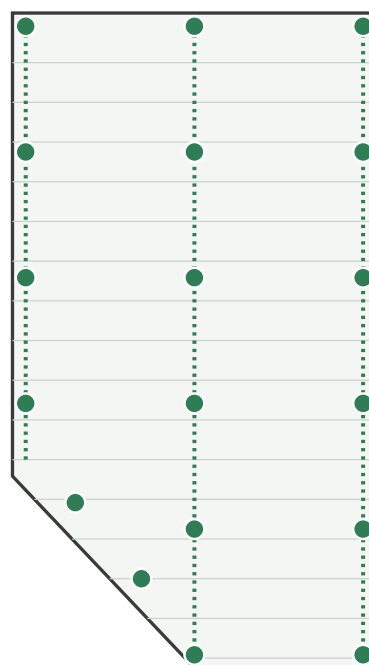
Ground-screw & bearer layout

Workshop 1 - 6.0 x 4.0 m - 15 ground screws



- Ground screw - 33 total (buy ~36 with spares)
- 5x2 bearer on screws - joists hung FLUSH both faces
- 5x2 joists @ 400 c/c, simple spans <= 2.0 m

Workshop 2 - ~4.0 x 7.2 m - 18 ground screws



clipped corner = angled door wall

On site / before you order

- **Set the bearers dead level first** on the adjustable heads — the flush joists reference off them, so any error shows in the deck.
- **Hanger nails, not screws** in the round holes (unless the hanger is marked for screws); fill every hole. That is the ~2000-nail line item — a palm nailer is worth it.
- **Heavy floor load?** Simple-span flush joists are weaker than the on-top continuous version — tighten bearers to 1.5 m (more hangers + a few more screws) or step up to 6x2.
- **Confirm Workshop 2's exact shape** so the WS2 hanger and screw counts can be firmed up.

Indicative take-off for planning only — not a structural design. Verify spans, hanger type, screw model, length and spacing against the manufacturers' load tables and your actual ground conditions before building.